ALI LAKHANI

✉️ a6lakhan@uwaterloo.ca 🔗 linkedin.com/in/ali-lakhani-a676b7175/ 🌐 Portfolio

Skills Summary

Mechanical: SolidWorks, Fusion360, Inventor, AutoCAD, Catia, DFM/DFA, GD&T, Tolerance Analysis
Hardware: Altium, KiCad, STM32, ESP32, Arduino, Adafruit, Sensor Integration (I^2C/SPI/UART)
Analysis/Testing: SimulationX, Ansys, FEA, Statistical Analysis, MATLAB and Python scripting
Software: Python, embedded C, C++, JavaScript (React, Node.js, TypeScript), SQL, VHDL, PLC

Education

University of Waterloo — **September 2023 - April 2028**
B.A.Sc - Honours Mechatronics Engineering, Co-op — *Waterloo, Ontario*

Experience

Vitruvius Venture Studio: Aerocardia — **September 2025 - December 2025**
Mechatronics Engineering Intern — *Montreal, Quebec*

- Designed a full mechanical enclosure system including a breathing tube (modelled similarly to a Venturi tube), mouthpiece, and multi-part shell housing all internal sensors and PCBs using **SolidWorks**, applying **DFM/DFA** to ensure manufacturability, simplified geometry, and fast assembly.
- Developed and deployed embedded firmware for a multi-sensor system, designing custom modular PCBs (including a wireless-charging-compatible PPG board) in **Altium** and integrating 7+ sensors controlled by an **ESP32**-S3 into a mechanically constrained enclosure, managing timing, power, and data integrity across the full electromechanical stack.
- Designed and validated mechanical enclosures and flow-critical components using **first-principles analysis** and iterative testing to ensure repeatable performance under real-world operating conditions.
- Engineered a high-resolution air volume-flow meter based on the Venturi effect, achieving 0.01 g/s resolution and reducing turbulent noise to within 1 Pa; processing 468 000 + data points in **MATLAB** and **Python** to derive an accurate real-time volume-flow.

Volt Carbon Technologies — **January 2025 - March 2025**
Manufacturing Engineering Intern — *Guelph, Ontario*

- Designed and executed performance and reliability tests for proprietary lithium-ion cells, analyzing efficiency, degradation trends, and failure behavior under real-world operating conditions.
- Modified and tested air classification chamber for graphite extraction from crushed ore; designed components in **Catia**
- Designed custom test fixtures and diagnostic setups to assess system performance, allowing for rapid iteration and easy root-cause analysis.
- Applied **DFM/DFA** on new process equipment, including magnetic separation funnels for iron removal and battery operation testers.
- Built **Python** automation tool to extract, clean, and visualize lab data from **TGA** and **XRF** systems, enhancing data accessibility and reporting efficiency

West Park Healthcare Center — **July 2023 - August 2023**
R&D Engineering Intern — *Toronto, Ontario*

- Engineered mouth-controlled gaming controller using **Arduino** and two custom PCBs in **KiCad**; integrated moisture-resistant pressure sensors to enhance usability for quadriplegic patients, increasing engagement by 40%

Projects/Extracurricular Experience

Vision Guided Autonomous Disk Launcher — **Link**
Mechanical Design Lead

- Designed and analyzed a dual-flywheel disk launcher and structural assembly in **SolidWorks**, integrating motor selection, flywheel geometry, Formlabs Flex 80A traction interfaces, and load-bearing components (housing, mounts, guide channel, magazine, safety enclosure) to achieve precise alignment, serviceability, and repeatable launch performance.
- Integrated and tuned dual ESP32-S3 motor controllers for high-speed flywheel actuation and pan-tilt control, managing real-time control loops, timing constraints, and safe bring-up of electromechanical hardware.
- Performed torque and inertia calculations to size pan-tilt actuators and determine acceleration limits, ensuring stable tracking without overshoot or mechanical resonance.
- Designed mechanical interfaces for a Raspberry Pi vision system and dual **ESP32**-S3 motor controllers, enabling reliable integration of perception, actuation, and safety hardware.
- Conducted structural **FEA** on launcher and turret components to validate stiffness, stress limits, and safety factors under dynamic loading.

VEX Competitive Robotics Team
Design Team Lead

- Designed competitive robot assemblies in **AutoCAD**, including a double reverse four-bar lift, x-drive base with omnidirectional wheels, and modular intake/outtake systems
- Applied **DFM principles** to optimize component alignment and precision, improving build consistency and competition performance
- Conducted **FEA** on motor mounts and structural elements to assess heat dissipation and material stress under extended operation
- Achieved 3rd place at the International Skills VEX Robotics Competition, demonstrating advanced design, CAD, and team coordination

Control Systems and Digital Logic Project Design
Project

- Implemented digital logic on Intel DE10-Lite **FPGA** using **VHDL** and **PLC** ladder logic in Siemens TIA Portal, designing a 3-bit synchronous counter and state machine; constructed state diagrams and validated transitions through simulation and real-time debugging.